

Juvenile Red King Crab × Ocean Acidification: an interim meta-analysis

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1. Goals

Three laboratory experiments at the Kodiak Laboratory have exposed juvenile red king crab (*Paralithodes camtschaticus*, RKC) to combinations of reduced pH and elevated temperature. Two have been published (Long et al. 2013, PLoS ONE 8(4): e60959; Swiney et al. 2017, ICES JMS 74: 1191–1200) and one was conducted in 2024–25 and remains unpublished. This document reports the first joint analysis of the three datasets, with two specific goals:

- (i) Bring the raw survival and growth (wet mass) records from all three experiments into a common, audit-ready structure that can be extended with future experiments.
- (ii) Quantify the effects of mean pH and mean temperature on daily mortality, contrasting an analysis restricted to the two published experiments with one that adds the 2024–25 data. The contrast tests whether including the new experiment substantively changes the inferred stressor effects, and surfaces any baseline differences between studies.

2. Methods

2.1 Data sources and preparation

Raw data from the three experiments were obtained as Excel workbooks from the Kodiak Laboratory archive. All sheets were converted to CSV; workbooks with multiple data-bearing tabs produced one CSV per tab. A single script (`scripts/data_wrangling.R`) ingests the CSVs and produces three harmonised tables: a daily-survival panel, an individual-level wet-mass panel, and a treatment-conditions table that carries the realised mean pH and mean temperature for each tank.

Realised conditions were computed from the daily pH and temperature records: the long-format 33529_RACE_2012-2013 file for 2012, and the wide-format NEW Juv RKC 2024 pH and Temp file for 2024. The 2010 experiment has no daily file in the archive; means and SDs reported in Long et al. 2013 (Table 1) are used in their place. All conditions are summarised in Table 1.

Experiment	pH treat.	Temp treat.	Mean pH (SD)	Mean temp °C (SD)
2010-2011	ambient	ambient	8.040 (0.040)	9.1 (2.0)
2010-2011	pH 7.8	ambient	7.802 (0.025)	9.1 (2.0)
2010-2011	pH 7.5	ambient	7.503 (0.040)	9.1 (2.0)
2012-2013	ambient	ambient	7.99 (0.03)	8.38 (1.62)
2012-2013	ambient	+2C	7.96 (0.04)	10.2 (1.49)
2012-2013	ambient	+4C	7.94 (0.04)	12.2 (1.63)
2012-2013	pH 7.8	ambient	7.79 (0.02)	8.55 (1.49)

2012-2013	pH 7.8	+2C	7.76 (0.02)	10.4 (1.49)
2012-2013	pH 7.8	+4C	7.74 (0.01)	12.3 (1.62)
2024-2025	ambient	ambient	7.88 (0.03)	8.53 (2.08)
2024-2025	ambient	+3C	7.88 (0.06)	10.3 (2.06)
2024-2025	pH 7.55	ambient	7.56 (0.03)	7.43 (1.92)
2024-2025	pH 7.55	+3C	7.54 (0.03)	10.4 (2.09)
2024-2025	pH 7.65	ambient	7.67 (0.03)	7.42 (1.92)
2024-2025	pH 7.65	+3C	7.65 (0.03)	10.2 (2.06)
2024-2025	pH 7.85	ambient	7.88 (0.05)	7.41 (1.91)
2024-2025	pH 7.85	+3C	7.85 (0.10)	10.3 (2.25)

Table 1. Realised treatment conditions. 2012 and 2024 values are computed from daily records; 2010 values are taken from Long et al. 2013 Table 1 because no daily file is present in the data folder.

2.2 Data-entry corrections

Audit of the 2024–25 worksheet uncovered thirteen data-entry anomalies, ranging from year typos (e.g. 2025-12-26 entered for a 4th-molt date that fell between two valid 2024 entries) to misplaced values, annotated dates, and crabs flagged for exclusion. The proposed corrections were documented in documents/2024-2025_data_issues_for_review.docx and reviewed by the original data holder (C. Long, NOAA AFSC) via track changes. Confirmed corrections are applied as documented `fix_cell_24()` calls in the wrangling script, keyed by (tub, cell), so the source spreadsheet remains unchanged and the corrections are auditable.

Two corrections are non-trivial. First, the ambient/ambient tub in 2024 (Tub 7) experienced a flow-cut event on 2024-12-16 that killed 14 of 15 crabs simultaneously; this is documented in a marginal note in the source sheet. Because those deaths are an experimental artefact, the ambient/ambient survival series is censored at the day before the failure (day 117). The one earlier death in that tub is retained as a real mortality event. Second, two 2024 crabs (Tub 5 Cell 24 and Tub 7 Cell 26) were flagged as “missing, not dead, remove from analysis”; they are dropped from both survival and growth analyses, reducing the starting count for those tanks by one.

2.3 Validation against published results

Before extending the analysis, the cleaned data were checked against the values reported in the two source papers. Long et al. 2013 fit a constant-hazard mortality model ($p_{\text{alive}} = \exp(-r \cdot t)$) by maximum likelihood and an exponential wet-mass growth model ($WM = a \cdot \exp(b \cdot t)$) with a random crab intercept on a . Refitting both models on the cleaned data reproduces the published coefficients to within rounding (mortality rates within 1%, wet-mass a exactly, b within 4%). Swiney et al. 2017 reported the relative effects of pH and temperature on mortality and the rank order of survival across treatments; both reproduce exactly (see scripts/check_against_papers.R). The cleaned data are therefore taken to be a faithful representation of the original experiments.

2.4 Survival model

Daily counts of live crabs were modelled with a Bayesian binomial hazard formulation that treats each tank as having a constant per-day mortality rate r_i :

$$\begin{aligned}
n_{\text{died}}_{i,t} \mid n_{\text{initial}}_i &\sim \text{Binomial}(n_{\text{initial}}_i, 1 - \exp(-r_i \cdot t)) \\
\log(r_i) &= \beta_0 + \beta_{\text{exp}} \cdot \text{exp}_i + \beta_{\text{pH}} \cdot \text{pH}_{c,i} + \beta_T \cdot \text{temp}_{c,i} + \beta_{\text{int}} \cdot \text{pH}_{c,i} \cdot \text{temp}_{c,i} + u_i \\
u_i &\sim \text{Normal}(0, \sigma_{\text{tank}})
\end{aligned}$$

Centering values were $\text{pH}_c = \text{pH} - 8.0$ and $\text{temp}_c = \text{temp} - 10^\circ\text{C}$. *exp* is a categorical for experiment (2010-11 reference, 2012-13 and 2024-25 dummies in the all-three model). u_i is a tank-level random intercept that absorbs treatment-level variation the linear predictor does not explain; with one tank per treatment, this is the closest available analogue to a biological replicate at the treatment level.

Priors were weakly informative: $\beta_0 \sim \text{Normal}(-5, 2)$ (spanning published mortality rates 0.001–0.05/day), $\beta_j \sim \text{Normal}(0, 5)$ for all slopes, and $\sigma_{\text{tank}} \sim \text{Exponential}(1)$. The likelihood was fit using the complementary log-log link with an offset of $\log(\text{day})$, which produces the constant-hazard parameterisation exactly: with the cloglog link and offset $\log(t)=0$, the linear predictor equals $\log(r)$.

Models were fit in R 4.6 with brms 2.23 calling Stan via rstan, four chains of 3,000 iterations (1,500 warmup), adapt_delta=0.95, and manual initialisation anchoring the intercept near -5 to avoid cloglog-init pathologies. Two model versions were fit, differing only in the experiments included: **Model A** uses the two published experiments (9 tanks, 1,585 daily observations); **Model B** adds 2024–25 (17 tanks, 3,774 observations).

3. Results

3.1 Model A: published experiments only

All chains converged (max R-hat = 1.004; minimum bulk ESS $\approx$ 1,355). The fitted per-tank rates reproduce the values reported in the two source papers to within rounding (Table 2): for the 2010 experiment specifically, the posterior medians match Long et al.'s ML estimates to the third decimal.

Tank	Median r (day ⁻¹)	95% CI
2010-2011 ambient ambient	0.00234	0.00221 – 0.00248
2010-2011 pH 7.8 ambient	0.00468	0.00447 – 0.00489
2010-2011 pH 7.5 ambient	0.0251	0.0238 – 0.0264
2012-2013 ambient ambient	0.00336	0.00318 – 0.00354
2012-2013 ambient +2C	0.00696	0.00666 – 0.00726
2012-2013 ambient +4C	0.00499	0.00475 – 0.00523
2012-2013 pH 7.8 ambient	0.00612	0.00587 – 0.00639
2012-2013 pH 7.8 +2C	0.00452	0.00430 – 0.00474
2012-2013 pH 7.8 +4C	0.0112	0.0108 – 0.0116

Table 2. Model A posterior per-tank mortality rates (constant per-day hazard).

Fixed-effect posteriors (log mortality-rate scale, Table 3) show a strong negative pH effect; the credible interval narrowly excludes zero. The temperature main effect and the pH $\times$ temperature interaction are not credibly distinguishable from zero in this dataset alone. There is no detectable baseline difference between the 2010 and 2012 experiments.

Parameter	Median	95% CI
Intercept (log r at pH 8.0, 10°C, 2010)	-5.67	-6.71 – -4.62
Experiment 2012 vs 2010	0.08	-0.98 – 1.13
pH (per unit; centred at 8.0)	-3.10	-5.90 – 0.06
Temperature (per °C; centred at 10°C)	0.16	-0.38 – 0.69
pH $\times$ Temperature interaction	0.52	-2.05 – 3.25
Tank-level SD	0.58	0.29 – 1.25

Table 3. Model A fixed-effect and tank-SD posteriors on the log mortality-rate scale.

Fitted survival vs observations per tank

Posterior median $\pm$ 95% CI from binomial hazard model

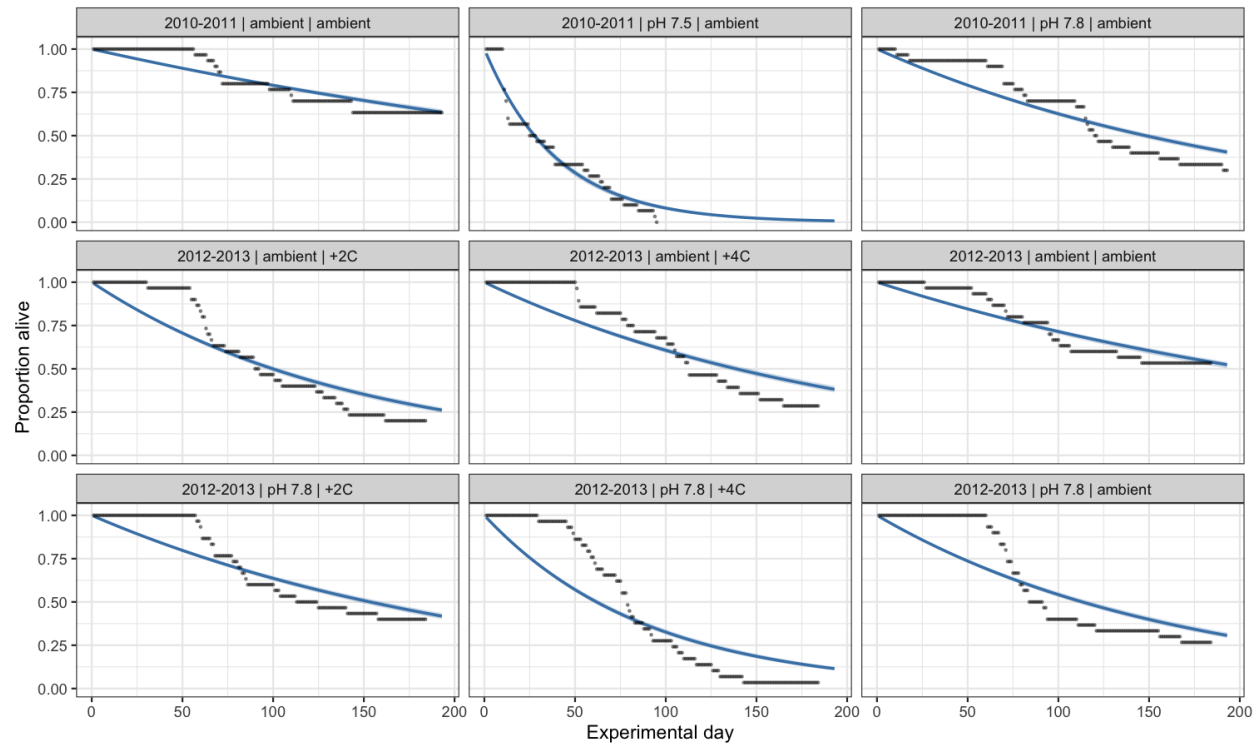


Figure 1. Model A: posterior median survival (blue line) with 95% credible interval (band) per tank, overlaid on observed daily proportions alive (points).

Posterior distributions of fixed effects + tank SD

Bayesian binomial hazard model, published RKC OA experiments

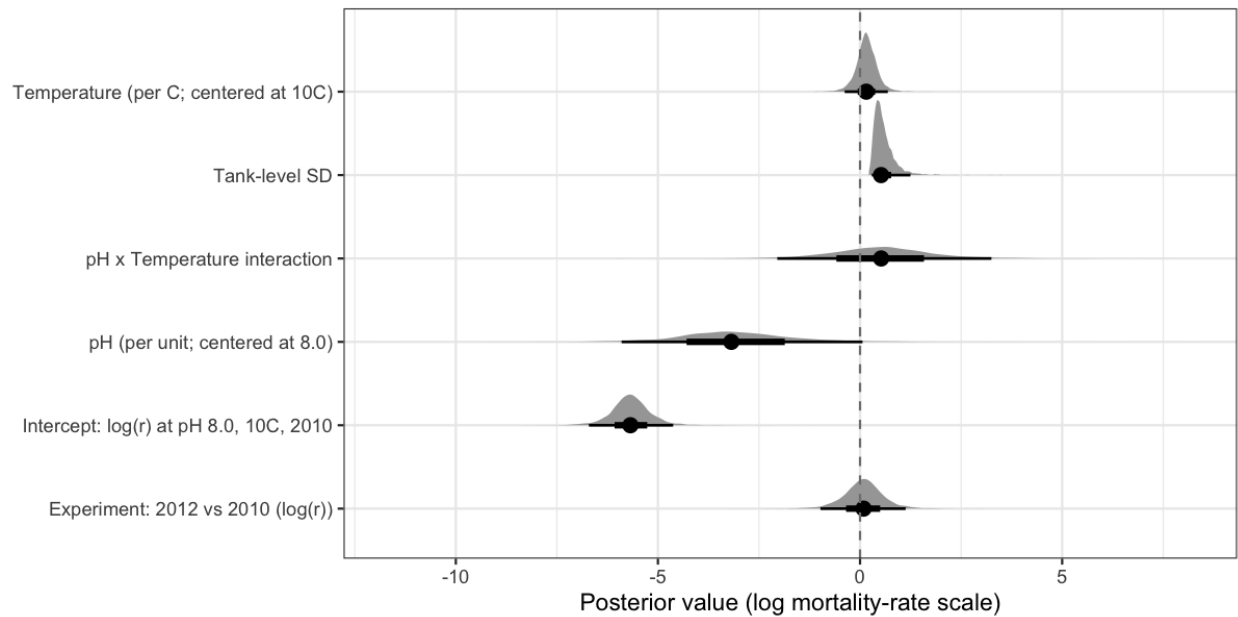


Figure 2. Model A: posterior distributions of fixed effects and the tank-level standard deviation.

Posterior median daily mortality rate over pH x temp

Population-level (random effect dropped); observed treatments shown as circles

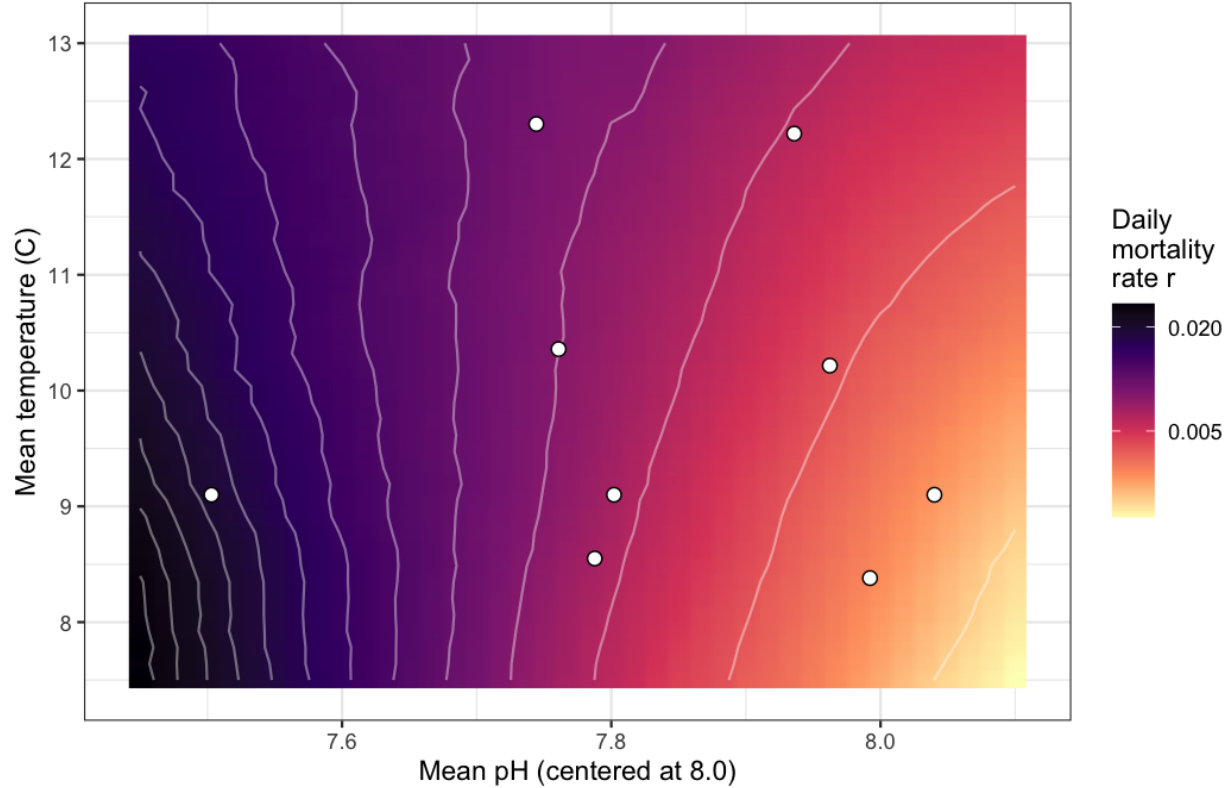


Figure 3. Model A: posterior median daily mortality rate over the realised pH $\times$ temperature space (random effect dropped). Observed treatments overlaid as circles. The near-vertical contours reflect the dominance of pH and the uninformative interaction.

3.2 Model B: all three experiments

Model B uses the same likelihood, prior, and tank random effect as Model A, with the experiment factor extended to three levels. All chains converged (max R-hat = 1.001; minimum bulk ESS $\approx$ 1,562).

The headline result is a large and credibly negative baseline shift for the 2024–25 experiment relative to 2010 (Table 4): $\beta_{\text{exp 2024 vs 2010}} = -2.21 [-3.41, -1.01]$, corresponding to an approximately 9× lower baseline mortality rate after accounting for pH and temperature. The pH main effect is preserved (still strongly negative on the log-rate scale) but attenuated relative to Model A; the temperature and interaction posteriors remain centred near zero with wide credible intervals. The tank-level SD increases from 0.58 to 0.83, reflecting additional between-tank variation in the expanded set.

Parameter	Median	95% CI
Intercept (log r at pH 8.0, 10°C, 2010)	−5.56	−6.76 – −4.35
Experiment 2012 vs 2010	0.04	−1.24 – 1.32
Experiment 2024 vs 2010	−2.21	−3.41 – −1.01
pH (per unit; centred at 8.0)	−2.56	−5.47 – 0.53
Temperature (per °C; centred at 10°C)	−0.07	−0.62 – 0.50
pH × Temperature interaction	−0.55	−2.51 – 1.51
Tank-level SD	0.83	0.55 – 1.28

Table 4. Model B fixed-effect and tank-SD posteriors.

The ranking of posterior per-tank rates (Table 5) puts every 2024–25 tank below the 2010 and 2012 controls. This is not a censoring artefact: only the ambient/ambient 2024 tank is censored (at day 117 for the documented flow failure); the other seven 2024 tanks ran the full ~297 days. The baseline shift is therefore a real, between-experiment difference in mortality after adjusting for realised pH and temperature.

Tank	Median r (day ^{−1})	95% CI
2024-25 ambient ambient	0.00023	0.00015 – 0.00034
2024-25 pH 7.85 +3C	0.00025	0.00021 – 0.00029
2024-25 pH 7.55 ambient	0.00039	0.00034 – 0.00043
2024-25 pH 7.65 +3C	0.00105	0.00097 – 0.00114
2024-25 pH 7.65 ambient	0.00113	0.00105 – 0.00123
2024-25 pH 7.55 +3C	0.00128	0.00119 – 0.00138
2024-25 ambient +3C	0.00137	0.00127 – 0.00147
2024-25 pH 7.85 ambient	0.00212	0.00200 – 0.00225
2010-11 ambient ambient	0.00234	0.00221 – 0.00248
2012-13 ambient ambient	0.00336	0.00318 – 0.00354
2012-13 pH 7.8 +2C	0.00451	0.00430 – 0.00473

2010-11 pH 7.8 ambient	0.00468	0.00446 – 0.00489
2012-13 ambient +4C	0.00498	0.00475 – 0.00522
2012-13 pH 7.8 ambient	0.00612	0.00587 – 0.00639
2012-13 ambient +2C	0.00696	0.00668 – 0.00726
2012-13 pH 7.8 +4C	0.0112	0.0108 – 0.0117
2010-11 pH 7.5 ambient	0.0250	0.0238 – 0.0263

Table 5. Model B posterior per-tank mortality rates, ranked from lowest to highest.

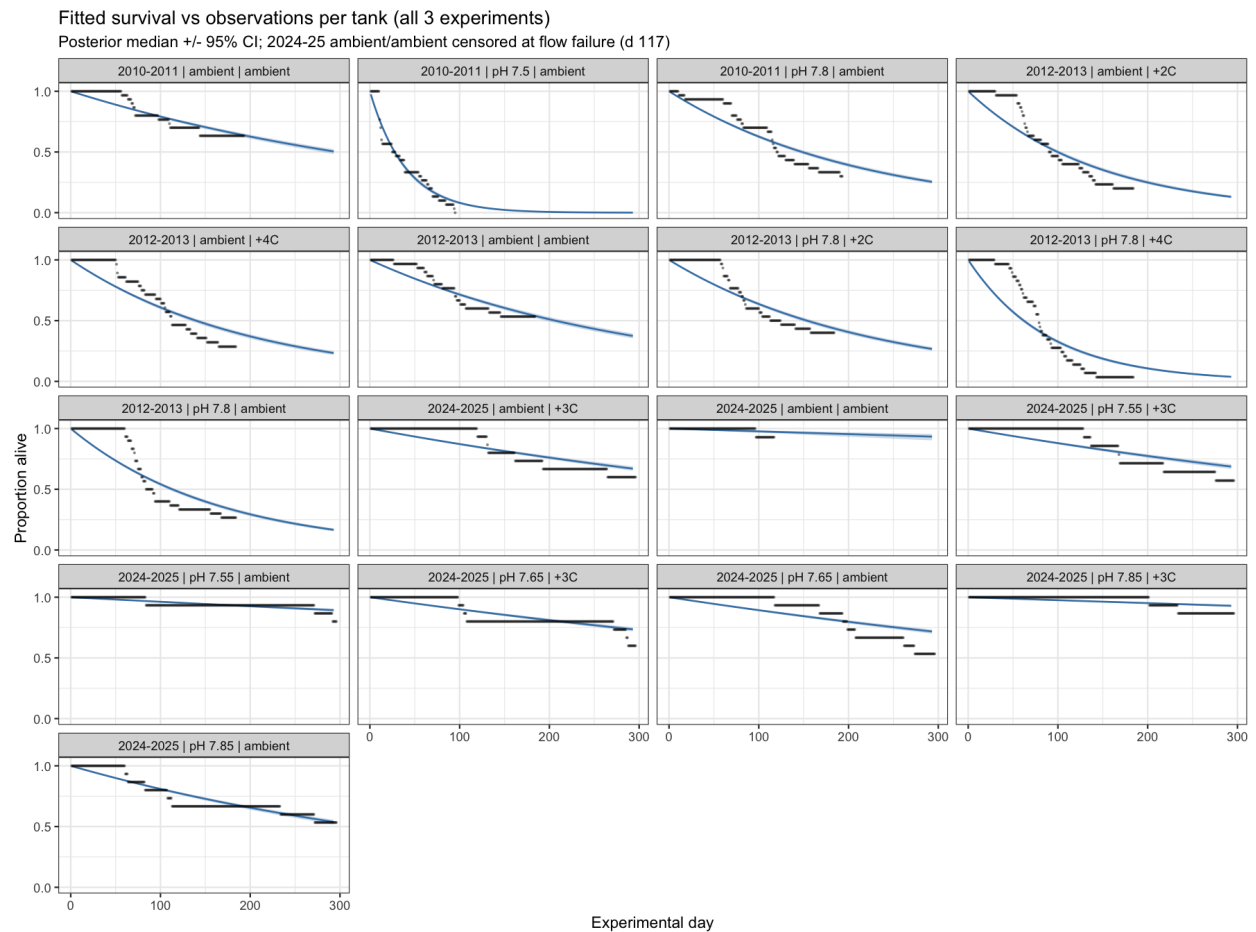


Figure 4. Model B: observed and fitted survival per tank, all three experiments. The 2024–25 panels (bottom rows) sit well above the 2010 and 2012 panels at comparable days, reflecting the lower baseline mortality. The ambient/ambient 2024 panel ends at day 117 by design (flow-failure censoring).

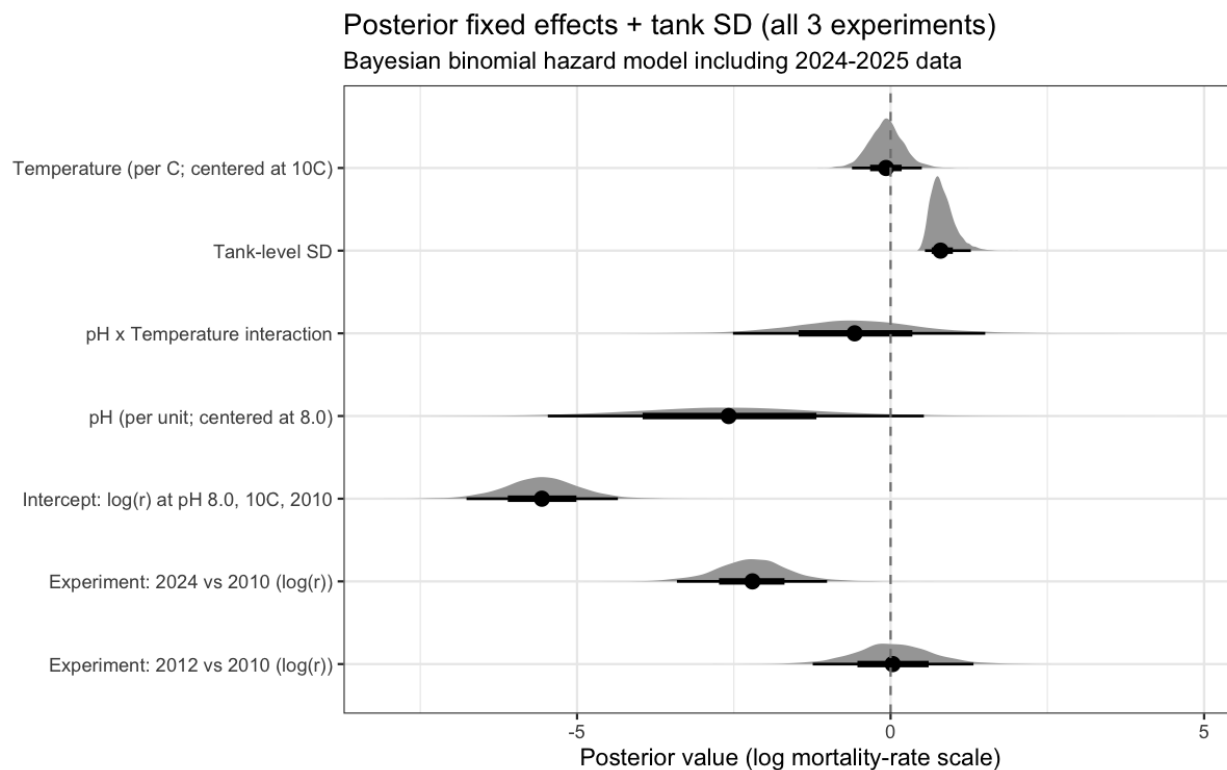


Figure 5. Model B: posterior distributions of fixed effects and the tank-level standard deviation.

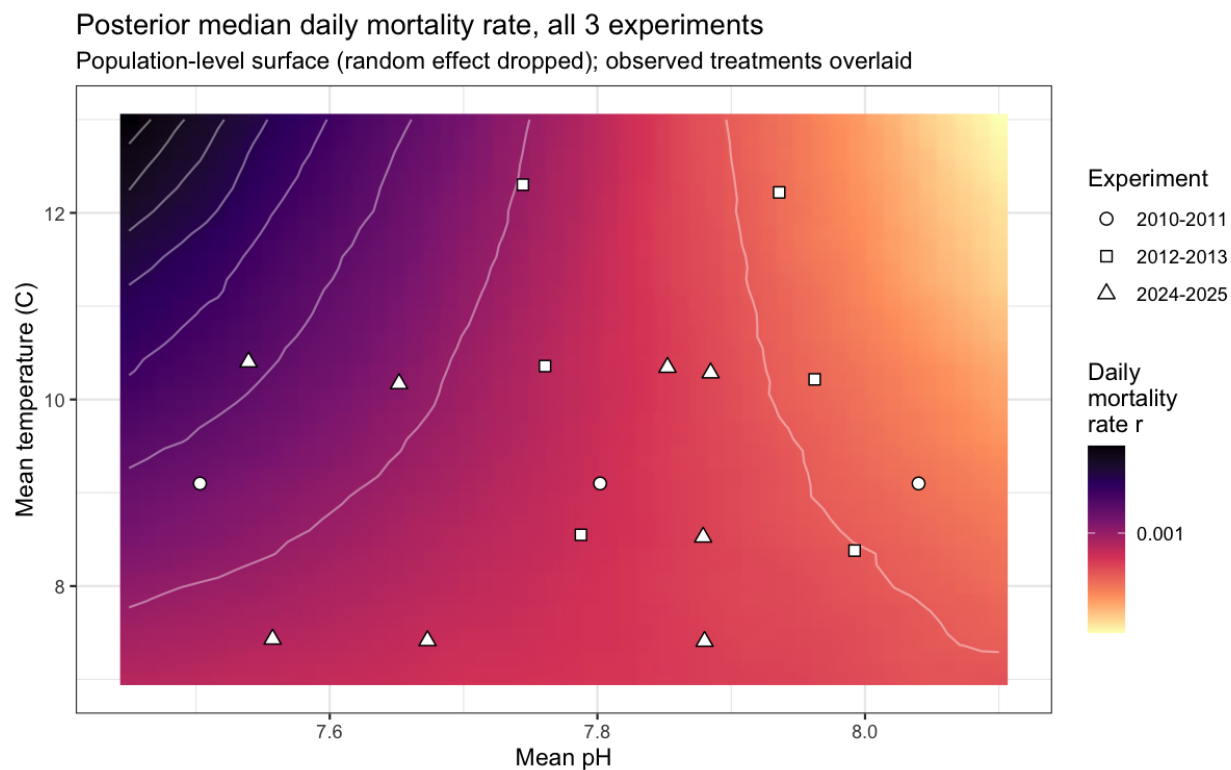


Figure 6. Model B: posterior median daily mortality rate over the expanded pH $\times$ temperature space. Observed treatments overlaid by experiment (circle = 2010-11, square = 2012-13, triangle = 2024-25).

4. Discussion

4.1 Comparison of Model A and Model B

Three points are worth highlighting in the contrast between the published-only and the all-three-experiment fits:

The pH signal is robust. The pH main effect remains strongly negative under both designs (Model A: $\beta_{\text{pH}} = -3.10$; Model B: -2.56), and the relative ordering of treatments within each experiment is preserved. The attenuation in Model B is consistent with the lower 2024 baseline pulling the regression line up at the low-pH end of the covariate range, but the qualitative conclusion that pH drives mortality is unchanged.

The temperature signal remains weak. Neither model identifies a credible temperature main effect or pH $\times$ temperature interaction. The temperature axis has been reasonably well sampled (7.4–12.3°C across realised tank means), but the limited between-tank replication and the experiment-level confounds limit what can be said about a temperature signal that is largely identified from variation within a single experiment (2012 has all the warm treatments; 2024 has +3°C but at a different baseline).

The 2024–25 baseline is conspicuously low. Model B places every 2024 tank below the 2010 and 2012 control tanks on the rate scale. Some likely contributors:

- **Cohort and life-history stage.** The 2024 crabs were young-of-the-year reared in the laboratory from larvae produced by a single ovigerous female, whereas 2010 and 2012 used field-collected juveniles. Genetic and maternal effects may have produced a more robust cohort.
- **Husbandry differences.** The diet recipe, feeding schedule, and tank geometry have evolved across the three studies. The 2024 protocol incorporated astaxanthin supplementation, which the earlier experiments did not.
- **Water-source and seasonal effects.** 2010 and 2012 experiments ran summer to early winter; 2024 ran August to June. Source-water temperature, salinity, and pH have all changed over the intervening decade.

Discriminating among these mechanisms is beyond the scope of the data assembled here; resolving the baseline shift will require either explicit covariate data (cohort source, diet composition, weight at start) or replicate experiments designed to isolate one factor. For the immediate purpose of the meta-analysis the cleanest interpretation is that the experiment-level intercept absorbs an unmeasured combination of these factors, and that the pH and temperature slopes are conditional on that baseline.

4.2 Next steps

Three immediate analytical extensions follow naturally from the current state of the work:

(a) Growth analysis. Wet-mass data are combined and harmonised across the three experiments (8–15 measurements per crab in 2012 and 2024; 3–5 in 2010). A non-linear growth model with crab-level random intercepts is the natural complement to the survival analysis.

(b) Sensitivity of the survival inference. The constant-hazard assumption is a strong one; the 2010 pH 7.5 cohort shows a delayed mortality cliff that the constant-rate model captures only in aggregate. A Weibull or piecewise-constant hazard would test whether the conclusions about pH and temperature effects survive relaxing the constant-rate assumption.

(c) Investigation of the 2024 baseline shift. If covariates describing cohort source, diet, and pre-experiment husbandry are available from laboratory notebooks, they could be added as fixed effects in place of the experiment dummy. If not, the contrast should be flagged in any manuscript as a limitation that constrains extrapolation to other crab populations.

5. Reproducibility

All steps are reproducible from the repository `juvenile-RKC-0A-experiments`:

- `scripts/data_wrangling.R` — ingest, document corrections, and write the three harmonised tables to `data/combined/`.
- `scripts/check_against_papers.R` — refit the published models on the cleaned data and compare to the values reported in Long et al. 2013 and Swiney et al. 2017.
- `scripts/survival.R` — fit Models A and B in brms, save the fits to `output/models/`, and produce the figures in `output/figures/`.
- `documents/session_transcript.md` — complete conversational record of the analytical work that produced this document.